DC Circuits — Reference

prefixes

Symbol	Prefix	Power	Multiply by	In words
G	giga	10^{9}	1,000,000,000	billion
M	mega	10^{6}	1,000,000	million
k	kilo	10^{3}	1,000	thousand
—	(none)	10^{0}	1	one
m	milli	10^{-3}	0.001	thousandth
μ	micro	10^{-6}	0.000 001	millionth
n	nano	10^{-9}	0.000 000 001	billionth
p	pico	10^{-12}	0.000 000 000 001	trillionth

quantities and units

Quantity	Symbol	Unit	In base units
charge	q, Q	coulomb (C)	A·s
current	i, I	ampere (A)	C/s
voltage	v, V	volt (V)	J/C
power	p, P	watt (W)	J/s = V·A
energy	w, W, E	joule (J)	W·s
resistance	R	ohm (Ω)	V/A
conductance	G	siemens (S)	A/V = $1/\Omega$
time	t	second (s)	—

Lowercase (q, i, v, p, w) means the value can change with time. Uppercase (Q, I, V, P, W) means a constant.

constants and conversions

Item	Value
charge of one electron	$e = 1.602 \times 10^{-19}$ C
electrons in 1 C	6.242×10^{18}
minute, hour	1 min = 60 s, 1 h = 3600 s
ampere-hour	1 Ah = 3600 C, 1 mAh = 3.6 C
watt-hour	1 Wh = 3600 J
kilowatt-hour	1 kWh = 3.6×10^{6} J
specific heat of water	$c = 4186$ J/(kg·°C)
mass of water	1 L ≈ 1 kg, 1 fl oz ≈ 0.02957 kg
cents to dollars	divide by 100

formula sheet

1–3. Charge and current

charge → electrons	$n = \dfrac{\lvert Q \rvert}{e}$
electrons → charge	$Q = n e$
constant current	$I = \dfrac{Q}{t}, \quad Q = I t$
constant current, t_1 to t_2	$Q = I\,(t_2 - t_1)$
changing charge → current	$i = \dfrac{dq}{dt}$
changing current → charge	$q = \int_{t_1}^{t_2} i\,dt$

4. Voltage

voltage	$v = \dfrac{w}{q}$
energy to move a charge	$w = q v$

5–6. Power and energy

power	$p = v i$ (sign: section 5)
power at one instant t_1	$p(t_1) = v(t_1) \cdot i(t_1)$
constant power → energy	$W = P t$
changing power → energy	$w = \int_{t_1}^{t_2} p\,dt$
energy → power	$p = \dfrac{dw}{dt}$
conservation	$\sum p = 0$

7. Batteries and utility energy

power, voltage → current	$I = \dfrac{P}{V}$
Ah rating → Wh rating	$\text{Wh} = V \cdot \text{Ah}$
run time from Ah	$t = \dfrac{\text{Ah}}{I}$ (hours)
run time from Wh	$t = \dfrac{\text{Wh}}{P}$ (hours)
batteries an energy fills	$n = \dfrac{W}{W_{\text{one}}}$, round down

8. Heat and cost

heat	$E = c m \Delta T$
temperature change	$\Delta T = \dfrac{E}{c m}$
cost	cost = kWh · rate

10. Resistors

Ohm's law	$v = i R$
conductance	$G = \dfrac{1}{R}, \quad i = G v$
power in a resistor	$p = i^2 R = \dfrac{v^2}{R} = v i$
energy in a resistor, constant I	$W = I^2 R t$

11–17. Circuit laws

branches, nodes, loops	$b = l + n - 1$
KCL	$\sum i_{\text{in}} = \sum i_{\text{out}}$
KCL equations	$n - 1$ for n nodes
resistor current from node voltages	$i = \dfrac{v_a - v_b}{R}$
KVL	$\sum v = 0$ around a loop
voltage from a to b	v_{ab} = sum of drops along a path
series	$R_{eq} = R_1 + R_2 + \cdots$
parallel	$\dfrac{1}{R_{eq}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots$
two in parallel	$R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$
voltage divider (series)	$v_1 = \dfrac{R_1}{R_1 + R_2}\, v$
N-resistor voltage divider	$v_k = \dfrac{R_k}{R_1 + \cdots + R_N}\, v$
current divider (parallel)	$i_1 = \dfrac{R_2}{R_1 + R_2}\, i$
N-branch current divider	$i_k = \dfrac{G_k}{G_1 + \cdots + G_N}\, i$
$\Delta \to Y$	$R_a = \dfrac{R_{ab} R_{ca}}{\Sigma}, \quad \Sigma = R_{ab} + R_{bc} + R_{ca}$
$Y \to \Delta$	$R_{ab} = \dfrac{N}{R_c}, \quad N = R_a R_b + R_b R_c + R_c R_a$

calculus tables

Derivatives (a, ω, n, c are constants):

Function	Derivative
c	0
t^n	$n t^{n-1}$
e^{at}	$a e^{at}$
$\sin(\omega t)$	$\omega \cos(\omega t)$
$\cos(\omega t)$	$-\omega \sin(\omega t)$
$f \cdot g$ (product rule)	$f' g + f g'$

Integrals:

Function	Antiderivative
c	$c t$
t^n	$\dfrac{t^{n+1}}{n+1}$
e^{at}	$\dfrac{e^{at}}{a}$
$\sin(\omega t)$	$-\dfrac{\cos(\omega t)}{\omega}$
$\cos(\omega t)$	$\dfrac{\sin(\omega t)}{\omega}$

Definite integral: $\int_{t_1}^{t_2} f\,dt = F(t_2) - F(t_1)$. F is the antiderivative. Top

minus bottom.

Exponent rules:

$e^{at} \cdot e^{bt} = e^{(a+b)t}$ $\qquad$ $\left(e^{at}\right)^2 = e^{2at}$

$e^0 = 1$ $\qquad$ $e^{-ab} \to 0$ when $b \to \infty$ $(a > 0)$

Shortcuts for a decaying exponential $(a > 0)$:

Integral	Result
$\int_0^T Ae^{-at}\,dt$	$\frac{A}{a}\left(1 - e^{-aT}\right)$
$\int_0^\infty Ae^{-at}\,dt$	$\frac{A}{a}$
$\int_0^\infty At\,e^{-at}\,dt$	$\frac{A}{a^2}$

Use the shortcuts only to check. Section 3 shows the full steps.

which tool? — given → find

Given → find	Do this	
charge $Q \to$ electrons	$n = \lvert Q \rvert / e$	(1)
constant numbers I, $t \to Q$	multiply: $Q = It$	(2)
$q(t) \to i$	differentiate	(3)
$i(t) \to q$	integrate	(3)
w, $q \to v$	divide: $v = w/q$	(4)
v, $i \to p$	multiply, sign from the convention	(5)
$p(t) \to w$	integrate (constant: $W = Pt$)	(6)
$w(t) \to p$	differentiate	(6)
$v(t)$, $i(t) \to p$ at t_1	put t_1 in each, then multiply	(6)
$v(t)$, $i(t) \to$ total energy	multiply, expand, integrate	(6)
battery rating → run time	$t =$ Ah$/I$ or Wh$/P$	(7)
mass, $\Delta T \to$ heat	$E = cm\Delta T$	(8)
joules → cost	$\div\, 3.6 \times 10^6$, then $\times$ rate	(8)
two of v, i, $R \to$ the third	Ohm's law, one resistor	(10)
resistor → power absorbed	i^2R or v^2/R	(10)
currents at a node → last one	KCL	(12)
voltages in a loop → last one	KVL	(13)
voltage between two points	path rule	(13)
network → R_{eq}	series and parallel	(14)
target $R_{eq} \to$ missing R	formula, solve for R	(14)
series, total $v \to$ one part	voltage divider	(15)
divider with a load	combine the load first	(15)
parallel, total $i \to$ one branch	current divider	(16)
N parallel branches, total i	divider with G	(16)
not series, not parallel	$\Delta \leftrightarrow Y$	(17)

The number in parentheses is the concept section.

DC Circuits — Concepts

1. charge and electrons

Charge is a property of matter, measured in coulombs (C). An electron carries a negative charge of size $e = 1.602 \times 10^{-19}$ C.
Charge to electrons:

$$n = \frac{|Q|}{e} = \frac{|Q|}{1.602 \times 10^{-19}\ \text{C}}$$

Electrons to charge:

$$Q = ne \quad \text{(write a minus sign for electrons)}$$

Put Q in coulombs before you divide. A prefix left on Q makes the count wrong by a power of ten.

Worked: the number of electrons in 2 mC

1. CONVERT THE PREFIX
$2\ \text{mC} = 2 \times 10^{-3}\ \text{C}$ *(milli = 10^{-3})*

↓

2. WRITE THE FORMULA
$n = \dfrac{|Q|}{e}$

↓

3. SUBSTITUTE
$n = \dfrac{2 \times 10^{-3}}{1.602 \times 10^{-19}}$

↓

4. DIVIDE THE NUMBERS
$\dfrac{2}{1.602} = 1.248$

↓

5. DIVIDE THE POWERS OF TEN
$\dfrac{10^{-3}}{10^{-19}} = 10^{-3-(-19)} = 10^{16}$ *(subtract the exponents)*

↓

6. COMBINE
$n = 1.248 \times 10^{16}$ electrons

Worked: (reverse) the charge of 5×10^{15} electrons

1. WRITE THE FORMULA
$Q = ne$

↓

2. SUBSTITUTE
$Q = \left(5 \times 10^{15}\right)\left(1.602 \times 10^{-19}\right)\ \text{C}$

↓

3. MULTIPLY THE NUMBERS
$5 \cdot 1.602 = 8.01$

↓

4. MULTIPLY THE POWERS OF TEN
$10^{15} \cdot 10^{-19} = 10^{15+(-19)} = 10^{-4}$ *(add the exponents)*

↓

5. COMBINE AND ADD THE SIGN
$Q = -8.01 \times 10^{-4}\ \text{C} = -0.801\ \text{mC}$ *(electrons are negative)*

↓

6. CHECK
$\dfrac{8.01 \times 10^{-4}}{1.602 \times 10^{-19}} = 5 \times 10^{15}$ ✓

2. current: the constant case

Current is the rate of flow of charge, measured in amperes (A). $1\ \text{A} = 1\ \text{C/s}$.
Direct current (DC) is a constant current I.

Current from charge and time: $I = \dfrac{Q}{t}$
Charge from current and time: $Q = It$
Charge over a time range t_1 to t_2: $Q = I(t_2 - t_1)$
Time must be in seconds. A coulomb is an ampere times a second. Convert minutes and hours first.

Conventional current flows the other way from electrons. The current arrow shows the flow of positive charge. A negative current means the real flow goes against the arrow.

Worked: 0.5 A flows for 2 min. Find the charge

1. CONVERT THE TIME
$2\ \text{min} = 2 \cdot 60 = 120\ \text{s}$

↓

2. WRITE THE FORMULA
$Q = It$

↓

3. SUBSTITUTE AND MULTIPLY
$Q = 0.5\ \text{A} \cdot 120\ \text{s} = 60\ \text{C}$

Worked: (reverse) 90 C passes in 30 s. Find the current

1. WRITE THE FORMULA
$I = \dfrac{Q}{t}$

↓

2. SUBSTITUTE AND DIVIDE
$I = \dfrac{90\ \text{C}}{30\ \text{s}} = 3\ \text{A}$

↓

3. CHECK
$Q = It = 3 \cdot 30 = 90\ \text{C}$ ✓

Worked: 4 mA flows from $t = 2$ s to $t = 7$ s. Find the charge

1. LENGTH OF THE RANGE
$t_2 - t_1 = 7 - 2 = 5\ \text{s}$

↓

2. SUBSTITUTE AND MULTIPLY
$Q = 4\ \text{mA} \cdot 5\ \text{s} = 20\ \text{mC}$ *(mA·s = mC, so the prefix passes through)*

3. current: the changing case

If the charge changes with time, $q(t)$, the current is its rate of change. The current is a function too, $i(t)$.
Charge to current (differentiate):

$$i(t) = \frac{dq}{dt}$$

Current to charge (integrate):

$$q = \int_{t_1}^{t_2} i(t)\,dt$$

Which way? — charge or current

Given → find	Do this
$q(t) \to i(t)$	Differentiate $q(t)$.
$i(t) \to q$	Integrate $i(t)$ between the two times.
constant Q, I, t	Section 2 (multiply or divide).

If the problem gives charge and your answer has a higher power of t, you integrated. Go the other way.
Put a time into the function only after you differentiate. A number put into $q(t)$ first becomes a constant, and its derivative is 0.
Units: the prefix passes through. mC → mA. nC → nA. μA → μC.

Worked: (differentiate) $q(t) = 3t^2$ mC. Find $i(t)$

1. PICK THE DIRECTION
Given charge, find current $\Rightarrow$ differentiate.

↓

2. POWER RULE
$\frac{d}{dt}\left(t^n\right) = n\,t^{n-1}$, here $n = 2$

↓

3. APPLY IT
$\frac{d}{dt}\left(3t^2\right) = 3 \cdot 2 \cdot t^{2-1} = 6t$

↓

4. UNITS
$i(t) = 6t$ mA *(mC per second = mA)*

↓

5. CHECK: INTEGRATE BACK
$\int_0^t 6\tau\,d\tau = 3t^2$ mC ✓

Worked: (differentiate) $q(t) = 0.8\left(1 - e^{-0.5t}\right)$ nC. Find $i(t)$

1. DISTRIBUTE
$q(t) = 0.8 - 0.8e^{-0.5t}$ nC

↓

2. DERIVATIVE OF THE CONSTANT
$\frac{d}{dt}(0.8) = 0$

↓

3. DERIVATIVE OF THE EXPONENTIAL
Rule: $\frac{d}{dt}e^{at} = a\,e^{at}$, here $a = -0.5$.
$\frac{d}{dt}\left(-0.8e^{-0.5t}\right) = -0.8 \cdot (-0.5)\,e^{-0.5t} = 0.4e^{-0.5t}$

↓

4. ADD THE TERMS
$i(t) = 0 + 0.4e^{-0.5t} = 0.4e^{-0.5t}$ nA

↓

5. CHECK THE ENDS
$t = 0$: $e^0 = 1$, so $i = 0.4$ nA. $t \to \infty$: $i \to 0$ while q levels off at 0.8 nC. ✓

Worked: (product rule) $q(t) = 0.5t\cos(100\pi t)$ μC. Find $i(t)$

1. NAME THE TWO FACTORS
$f = 0.5t, \quad g = \cos(100\pi t)$

↓

2. DIFFERENTIATE f
$f' = 0.5$

↓

3. DIFFERENTIATE g
Rule: $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t), \quad \omega = 100\pi$.
$g' = -100\pi\sin(100\pi t)$

↓

4. FIRST TERM $f'g$
$0.5\cos(100\pi t)$

↓

5. SECOND TERM fg'
$0.5t \cdot \left(-100\pi\sin(100\pi t)\right) = -50\pi t\sin(100\pi t)$

↓

6. ADD
$i(t) = 0.5\cos(100\pi t) - 50\pi t\sin(100\pi t)$ μA

Worked: (integrate) $i(t) = 20e^{-0.5t}$ mA, from $t = 0$ to $t = 4$ s. Find q

1. SET UP THE INTEGRAL
$q = \int_0^4 20e^{-0.5t}\,dt$

↓

2. ANTIDERIVATIVE
Rule: $\int e^{at}\,dt = \frac{e^{at}}{a}, \quad a = -0.5$.
$F(t) = \frac{20}{-0.5}e^{-0.5t} = -40e^{-0.5t}$

↓

3. EVALUATE AT THE TOP ($t = 4$)
$F(4) = -40e^{-0.5\cdot 4} = -40e^{-2} = -40(0.1353) = -5.413$

↓

4. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -40e^0 = -40(1) = -40$ *($e^0 = 1$, not 0)*

↓

5. TOP MINUS BOTTOM
$q = F(4) - F(0) = -5.413 - (-40) = -5.413 + 40 = 34.59$ mC *(mA·s = mC)*

↓

6. CHECK WITH THE SHORTCUT
$\frac{A}{a}\left(1 - e^{-aT}\right) = \frac{20}{0.5}\left(1 - 0.1353\right) = 40(0.8647) = 34.59$ ✓

Worked: (integrate to ∞) $i(t) = 0.4e^{-0.5t}$ nA. Find the total charge

1. SET UP WITH A FINITE TOP b
$q = \lim_{b\to\infty} \int_0^b 0.4e^{-0.5t}\,dt$ *(you cannot put ∞ into a function, so use b)*

↓

2. ANTIDERIVATIVE
$F(t) = \frac{0.4}{-0.5}e^{-0.5t} = -0.8e^{-0.5t}$

↓

3. EVALUATE AT THE TOP ($t = b$)
$F(b) = -0.8e^{-0.5b}$

↓

4. LET $b \to \infty$
$e^{-0.5b} \to 0$, so $F(b) \to -0.8(0) = 0$

↓

5. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -0.8e^0 = -0.8(1) = -0.8$

↓

6. TOP MINUS BOTTOM
$q = 0 - (-0.8) = 0.8$ nC

↓

7. CHECK WITH THE SHORTCUT
$\frac{A}{a} = \frac{0.4}{0.5} = 0.8$ ✓ *(same as where $q(t)$ levels off two examples up)*

A total to ∞ exists only if $i(t) \to 0$. A constant or a sinusoid has no finite total.

4. voltage

Voltage is the energy needed to move one coulomb of charge between two points. It is measured in volts (V). 1 V $= 1$ J/C.

Voltage from energy and charge: $v = \frac{w}{q}$

Energy from charge and voltage: $w = q\,v$

Current flows *through* an element. Voltage is *across* an element, from its $+$ mark to its $-$ mark. $v_{ab} = -v_{ba}$.

Worked: move 2 C through 12 V. Find the energy

1. WRITE THE FORMULA
$w = q\,v$

↓

2. SUBSTITUTE AND MULTIPLY
$w = 2$ C $\cdot\, 12$ V $= 24$ J *(C·V = C·J/C = J)*

Worked: (reverse) 30 J moves 5 C. Find the voltage

1. WRITE THE FORMULA
$$v = \frac{w}{q}$$

↓

2. SUBSTITUTE AND DIVIDE
$$v = \frac{30\ \text{J}}{5\ \text{C}} = 6\ \text{V}$$

5. power and the passive sign convention

Power is the rate of energy transfer, measured in watts (W). $1\ \text{W} = 1\ \text{J/s}$.

$$p = v\,i$$

Passive sign convention: look at where the current arrow enters the element.
- The arrow enters the $+$ mark: $p = +v\,i$.
- The arrow enters the $-$ mark: $p = -v\,i$.

Then read the sign of the result:
- $p > 0$: the element **absorbs** power (a resistor, a battery that charges).
- $p < 0$: the element **delivers** power (a source).



Conservation of energy: in each circuit, $\sum p = 0$. The power delivered equals the power absorbed.
The sign comes from the arrow and the marks, not from the element type. Check the arrow against the $+$ mark before you multiply.

Worked: absorbs or delivers? Three cases

1. $v = 5$ V, $i = 2$ A **INTO THE $+$ MARK**
Formula: $p = +v\,i$. $p = +(5)(2) = +10$ W.
$p > 0 \Rightarrow$ absorbs 10 W.

↓

2. $v = 5$ V, $i = 2$ A **INTO THE $-$ MARK**
Formula: $p = -v\,i$. $p = -(5)(2) = -10$ W.
$p < 0 \Rightarrow$ delivers 10 W.

↓

3. $v = -3$ V, $i = 4$ A **INTO THE $+$ MARK**
Formula: $p = +v\,i$. $p = +(-3)(4) = -12$ W.
$p < 0 \Rightarrow$ delivers 12 W. *(the negative voltage flips the sign)*

Worked: conservation check

A 12 V source drives 2 A through one resistor. The current leaves the $+$ mark of the source and enters the $+$ mark of the resistor.

1. SOURCE
The arrow enters the $-$ mark. $p = -(12)(2) = -24$ W (delivers)

↓

2. RESISTOR
The arrow enters the $+$ mark. $p = +(12)(2) = +24$ W (absorbs)

↓

3. ADD
$\sum p = -24 + 24 = 0$ ✓

6. energy

Energy is the amount. Power is the rate. The pattern is the same as charge (amount) and current (rate).
Energy from constant power: $W = Pt$ (t in seconds gives joules)
Energy from changing power:

$$w = \int_{t_1}^{t_2} p(t)\,dt = \int_{t_1}^{t_2} v(t)\,i(t)\,dt$$

Power from energy: $p = \dfrac{dw}{dt}$
Power at one instant t_1: $p(t_1) = v(t_1) \cdot i(t_1)$

Which way? — energy or power

Given → find	Do this
constant P, $t \to W$	Multiply: $W = Pt$.
$p(t) \to w$	Integrate $p(t)$ between the two times.
$w(t) \to p$	Differentiate $w(t)$.
$v(t)$, $i(t) \to p(t_1)$	Put t_1 into each. Then multiply.
$v(t)$, $i(t) \to w$	Multiply the functions. Then integrate.

$\int v\,i\,dt$ is not $\left(\int v\,dt\right) \cdot \left(\int i\,dt\right)$. Multiply the functions first.
Units: V·mA = mW. mW·s = mJ.

Worked: (constant) a 12 V battery delivers 2 A for 1 h

1. POWER
$P = V\,I = 12\ \text{V} \cdot 2\ \text{A} = 24\ \text{W}$

↓

2. CONVERT THE TIME
$1\ \text{h} = 3600\ \text{s}$

↓

3. ENERGY
$W = Pt = 24\ \text{W} \cdot 3600\ \text{s} = 86{,}400\ \text{J} = 86.4\ \text{kJ}$

↓

4. SAME ENERGY IN Wh
$24\ \text{W} \cdot 1\ \text{h} = 24\ \text{Wh}$. Check: $24 \cdot 3600 = 86{,}400$ J ✓

Worked: (reverse) $w(t) = 5t^2$ J. Find the power at $t = 3$ s

1. PICK THE DIRECTION
Given energy, find power ⇒ differentiate.

↓

2. DIFFERENTIATE
$$p(t) = \frac{d}{dt}\left(5t^2\right) = 5 \cdot 2 \cdot t = 10t\ \text{W}$$

↓

3. PUT IN THE TIME
$p(3) = 10 \cdot 3 = 30\ \text{W}$

Worked: $v = 50\left(1 - e^{-0.5t}\right)$ V, $i = 4e^{-0.5t}$ mA. Power at $t = 1$ s

1. THE EXPONENTIAL AT $t = 1$
$e^{-0.5 \cdot 1} = e^{-0.5} = 0.6065$

↓

2. VOLTAGE AT $t = 1$
$v(1) = 50(1 - 0.6065) = 50(0.3935) = 19.67\ \text{V}$

↓

3. CURRENT AT $t = 1$
$i(1) = 4(0.6065) = 2.426\ \text{mA}$

↓

4. MULTIPLY
$p(1) = 19.67\ \text{V} \cdot 2.426\ \text{mA} = 47.7\ \text{mW}$ *(V·mA = mW)*

Worked: same device, total energy from 0 to ∞

1. SET UP
$w = \int_0^{\infty} v(t)\, i(t)\, dt = \int_0^{\infty} 50\left(1 - e^{-0.5t}\right) \cdot 4e^{-0.5t}\, dt$

↓

2. MULTIPLY THE CONSTANTS
$50 \cdot 4 = 200$, so $p(t) = 200\left(1 - e^{-0.5t}\right)e^{-0.5t}$ mW

↓

3. DISTRIBUTE $e^{-0.5t}$
$p(t) = 200\left(e^{-0.5t} - e^{-0.5t} \cdot e^{-0.5t}\right)$

↓

4. COMBINE THE EXPONENTS
$e^{-0.5t} \cdot e^{-0.5t} = e^{(-0.5-0.5)t} = e^{-t}$ *(add the exponents)*

↓

5. DISTRIBUTE THE 200
$p(t) = 200e^{-0.5t} - 200e^{-t}$ mW

↓

6. ANTIDERIVATIVE OF THE FIRST TERM
$\int 200e^{-0.5t}\, dt = \dfrac{200}{-0.5}e^{-0.5t} = -400e^{-0.5t}$

↓

7. ANTIDERIVATIVE OF THE SECOND TERM
$\int 200e^{-t}\, dt = \dfrac{200}{-1}e^{-t} = -200e^{-t}$

↓

8. FULL ANTIDERIVATIVE
$F(t) = -400e^{-0.5t} - \left(-200e^{-t}\right) = -400e^{-0.5t} + 200e^{-t}$ *(first minus second, as in step 5)*

↓

9. EVALUATE AT THE TOP ($t = b$)
$F(b) = -400e^{-0.5b} + 200e^{-b}$

↓

10. LET $b \to \infty$
$e^{-0.5b} \to 0$ and $e^{-b} \to 0$, so $F(b) \to -400(0) + 200(0) = 0$

↓

11. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -400e^{0} + 200e^{0} = -400(1) + 200(1) = -200$

↓

12. TOP MINUS BOTTOM
$w = 0 - (-200) = 200$ mJ *(mW·s = mJ)*

↓

13. CHECK WITH THE SHORTCUT $\frac{A}{a}$ ON EACH TERM
$\dfrac{200}{0.5} - \dfrac{200}{1} = 400 - 200 = 200$ ✓

7. battery and utility units: Ah, Wh, kWh

Batteries and utility bills use hours, not seconds.

- **Ampere-hour** (Ah) is charge: 1 Ah $= 1$ A $\cdot\ 3600$ s $= 3600$ C.
- **Watt-hour** (Wh) is energy: 1 Wh $= 1$ W $\cdot\ 3600$ s $= 3600$ J.
- **Kilowatt-hour** (kWh) is energy: 1 kWh $= 1000 \cdot 3600 = 3.6 \times 10^6$ J.

Read the unit of a battery rating. Ah or mAh is charge. Wh or kWh is energy.

Given → find	Do this
P, $V \to I$	$I = \dfrac{P}{V}$
Ah, $V \to$ Wh	Wh $= V \cdot$ Ah
Wh, $V \to$ Ah	Ah $= \dfrac{\text{Wh}}{V}$
Ah rating, $I \to$ run time	$t = \dfrac{\text{Ah}}{I}$ (hours)
Wh rating, $P \to$ run time	$t = \dfrac{\text{Wh}}{P}$ (hours)
Wh rating, V, $I \to$ run time	$P = V\,I$ first, then $t = \dfrac{\text{Wh}}{P}$
Wh → J, Ah → C	multiply by 3600
mAh → Ah	divide by 1000
energy $W \to$ batteries filled	$n = \dfrac{W}{W_{\text{one}}}$, same units, round down

Worked: a 2.5 Ah battery, load draws 500 mA. Run time

1. CONVERT THE CURRENT
500 mA $= 0.5$ A

↓

2. WRITE THE FORMULA
$t = \dfrac{\text{Ah}}{I}$

↓

3. SUBSTITUTE AND DIVIDE
$t = \dfrac{2.5 \text{ Ah}}{0.5 \text{ A}} = 5$ h

Worked: (Ah to Wh) the same battery at 3.7 V

1. WRITE THE FORMULA
Wh $= V \cdot$ Ah

↓

2. SUBSTITUTE AND MULTIPLY
Wh $= 3.7$ V $\cdot\ 2.5$ Ah $= 9.25$ Wh

↓

3. TO JOULES
$9.25 \cdot 3600 = 33{,}300$ J

Worked: a 12 V, 60 Wh battery, load draws 250 mA

1. CONVERT THE CURRENT
250 mA $= 0.25$ A

↓

2. LOAD POWER
$P = V\,I = 12$ V $\cdot\ 0.25$ A $= 3$ W

↓

3. RUN TIME
$t = \dfrac{\text{Wh}}{P} = \dfrac{60 \text{ Wh}}{3 \text{ W}} = 20$ h

↓

4. CHECK WITH CHARGE
Ah $= \dfrac{60}{12} = 5$ Ah. 0.25 A $\cdot\ 20$ h $= 5$ Ah ✓

Worked: a 240 V heater rated 1.2 kW, on for 15 min

1. CONVERT THE POWER
1.2 kW $= 1200$ W

↓

2. CURRENT
$I = \dfrac{P}{V} = \dfrac{1200}{240} = 5$ A

↓

3. CONVERT THE TIME
15 min $= 15 \cdot 60 = 900$ s

↓

4. ENERGY IN JOULES
$W = Pt = 1200 \cdot 900 = 1{,}080{,}000$ J $= 1.08$ MJ

↓

5. SAME ENERGY IN kWh
15 min $= 0.25$ h. 1.2 kW $\cdot\ 0.25$ h $= 0.3$ kWh

↓

6. CHECK
$0.3 \cdot 3.6 \times 10^6 = 1.08 \times 10^6$ J ✓

Worked: how many 3000 mAh, 3.7 V batteries does 1.08 MJ fill?

1. mAh TO Ah
$3000\ \text{mAh} = 3\ \text{Ah}$

↓

2. ENERGY OF ONE BATTERY IN Wh
$3.7\ \text{V} \cdot 3\ \text{Ah} = 11.1\ \text{Wh}$

↓

3. ENERGY OF ONE BATTERY IN J
$11.1 \cdot 3600 = 39{,}960\ \text{J}$

↓

4. DIVIDE
$$n = \frac{1{,}080{,}000}{39{,}960} = 27.03$$

↓

5. ROUND DOWN
27 full charges *(the 0.03 is not a full battery)*

Worked: a pulse of 8 kA for 20 μs across 100 MV

1. CONVERT EACH PREFIX
$I = 8\ \text{kA} = 8 \times 10^{3}\ \text{A}$.
$t = 20\ \mu\text{s} = 20 \times 10^{-6} = 2 \times 10^{-5}\ \text{s}$.
$V = 100\ \text{MV} = 100 \times 10^{6} = 10^{8}\ \text{V}$.

↓

2. CHARGE
$$Q = It = \left(8 \times 10^{3}\right)\left(2 \times 10^{-5}\right) = 16 \times 10^{-2} = 0.16\ \text{C}$$

↓

3. ELECTRONS
$$n = \frac{0.16}{1.602 \times 10^{-19}} = 0.0999 \times 10^{19} \approx 1.0 \times 10^{18}$$

↓

4. POWER
$$P = VI = \left(10^{8}\right)\left(8 \times 10^{3}\right) = 8 \times 10^{11}\ \text{W} = 800\ \text{GW}$$

↓

5. ENERGY
$$W = Pt = \left(8 \times 10^{11}\right)\left(2 \times 10^{-5}\right) = 16 \times 10^{6}\ \text{J} = 16\ \text{MJ}$$

↓

6. CHECK WITH $W = QV$
$0.16 \cdot 10^{8} = 1.6 \times 10^{7}\ \text{J} = 16\ \text{MJ}$ ✓

8. heat and energy cost

Heat to change the temperature of a mass:

$$E = cm\Delta T$$

- c: **specific heat**. Water: $c = 4186\ \text{J/(kg·°C)}$.
- m: mass in kg. Water: $1\ \text{L} \approx 1\ \text{kg}$. $1\ \text{fl oz} \approx 0.02957\ \text{kg}$.
- $\Delta T = T_{\text{final}} - T_{\text{initial}}$, in °C.

Temperature change from energy: $\Delta T = \dfrac{E}{cm}$

If all the electric energy becomes heat: $E_{\text{heat}} = W_{\text{electric}} = VIt$.

Cost: convert joules to kWh, then multiply by the rate.

$$\text{kWh} = \frac{\text{J}}{3.6 \times 10^{6}} \qquad \text{cost} = \text{kWh} \cdot \text{rate}$$

Round down a count of items. 191.1 cups means 191 full cups.

Worked: heat 0.5 kg of water from 20°C to 80°C at 15 cents/kWh

1. TEMPERATURE CHANGE
$\Delta T = 80 - 20 = 60°\text{C}$

↓

2. HEAT
$E = cm\Delta T = 4186 \cdot 0.5 \cdot 60 = 125{,}580\ \text{J}$

↓

3. JOULES TO kWh
$$\frac{125{,}580}{3.6 \times 10^{6}} = 0.03488\ \text{kWh}$$

↓

4. COST OF ONE
$0.03488\ \text{kWh} \cdot 15\ \text{cents/kWh} = 0.523\ \text{cents}$

↓

5. HOW MANY FOR $1
$\$1 = 100$ cents. $\dfrac{100}{0.523} = 191.1 \Rightarrow 191$

Worked: (reverse) 50 kJ goes into 0.4 kg of water. Find ΔT

1. CONVERT THE ENERGY
$50\ \text{kJ} = 50{,}000\ \text{J}$

↓

2. THE BOTTOM: cm
$4186 \cdot 0.4 = 1674.4\ \text{J/°C}$

↓

3. DIVIDE
$$\Delta T = \frac{50{,}000}{1674.4} = 29.9°\text{C}$$

Worked: a heater at 10 A, 120 V for 3 min, same water and rate

1. CONVERT THE TIME
$3\ \text{min} = 3 \cdot 60 = 180\ \text{s}$

↓

2. POWER
$P = VI = 120 \cdot 10 = 1200\ \text{W}$

↓

3. ENERGY
$W = Pt = 1200 \cdot 180 = 216{,}000\ \text{J}$

↓

4. HOW MANY BATCHES
$$\frac{216{,}000}{125{,}580} = 1.72 \Rightarrow 1\ \text{full batch}$$

↓

5. COST
$\dfrac{216{,}000}{3.6 \times 10^{6}} = 0.06\ \text{kWh}$. $0.06 \cdot 15 = 0.9$ cents

9. circuit elements

Active elements can deliver energy (sources). **Passive** elements cannot (resistors, capacitors, inductors).

- **Independent voltage source** (circle): fixed v. The current is whatever the circuit needs.
- **Independent current source** (circle with an arrow): fixed i. The voltage is whatever the circuit needs.
- **Dependent source** (diamond): its value is a gain times a voltage v_x or a current i_x somewhere else in the circuit.

Type	Name	Output	Gain unit
VCVS	voltage-controlled voltage source	$v = \mu v_x$	none
CCVS	current-controlled voltage source	$v = r i_x$	Ω
VCCS	voltage-controlled current source	$i = g v_x$	S
CCCS	current-controlled current source	$i = \beta i_x$	none

Two different ideal voltage sources in parallel, or two different ideal current sources in series, break KVL or KCL.

Worked: the value of a dependent source

1. READ THE TYPE
A diamond labeled $10\,i_x$ with $+$ and $-$ marks: a CCVS. Gain $r = 10\ \Omega$.

↓

2. FIND THE CONTROL
The circuit has $i_x = 0.5$ A.

↓

3. MULTIPLY
$v = 10\,\Omega \cdot 0.5\ \text{A} = 5\ \text{V}$ *($\Omega \cdot A = V$)*

10. Ohm's law and resistor power

Ohm's law:

$$v = iR \qquad i = \frac{v}{R} \qquad R = \frac{v}{i}$$

R is the **resistance** in ohms (Ω). **Conductance** $G = \frac{1}{R}$, in siemens (S). Then $i = Gv$.

Power absorbed by a resistor:

$$p = vi = i^2R = \frac{v^2}{R}$$

Energy absorbed by a resistor: constant current: $W = I^2Rt$. Changing current: $w = \int i^2R\,dt$.

A resistor always absorbs power ($p \geq 0$). All of it becomes heat.

- **Short circuit**: $R = 0$, so $v = 0$ for each current.
- **Open circuit**: $R = \infty$, so $i = 0$ for each voltage.

Ohm's law uses the passive sign convention. If the current arrow enters the $-$ mark, write $v = -iR$.

Which way? — Ohm's law or a definition?

Ohm's law connects v, i, and R of one resistor at the same instant. Use it only when you know two of the three. The definitions ($i = \frac{dq}{dt}$, $p = vi$) work for every element and need no R.

Known → find	Do this
$v, R \to i$	Ohm: $i = \frac{v}{R}$
$i, R \to v$	Ohm: $v = iR$
$v, i \to R$	Ohm: $R = \frac{v}{i}$
$i, R \to p$	$p = i^2R$
$v, R \to p$	$p = \frac{v^2}{R}$
$q(t) \to i$	Definition: $i = \frac{dq}{dt}$. No R.

Worked: 10 V across a 2 kΩ resistor. Find the current

1. CONVERT THE PREFIX
$2\ \text{k}\Omega = 2000\,\Omega$

↓

2. OHM'S LAW
$i = \frac{v}{R} = \frac{10}{2000} = 0.005\ \text{A} = 5\ \text{mA}$

Worked: (reverse) 3 mA through 4.7 kΩ. Find the voltage

1. CONVERT THE PREFIXES
$3\ \text{mA} = 0.003$ A. $4.7\ \text{k}\Omega = 4700\,\Omega$

↓

2. OHM'S LAW
$v = iR = 0.003 \cdot 4700 = 14.1$ V

Worked: power absorbed by a $50\,\Omega$ resistor that carries 0.2 A

1. WAY 1: $p = i^2R$
$i^2 = (0.2)^2 = 0.04$. $p = 0.04 \cdot 50 = 2$ W

↓

2. FIND v FOR THE OTHER WAYS
$v = iR = 0.2 \cdot 50 = 10$ V

↓

3. WAY 2: $p = \frac{v^2}{R}$
$v^2 = 10^2 = 100$. $p = \frac{100}{50} = 2$ W ✓

↓

4. WAY 3: $p = vi$
$p = 10 \cdot 0.2 = 2$ W ✓ *(all three ways agree)*

Worked: energy absorbed by $100\,\Omega$ carrying 0.5 A for 10 min

1. POWER
$P = I^2R = (0.5)^2 \cdot 100 = 0.25 \cdot 100 = 25$ W

↓

2. CONVERT THE TIME
$10\ \text{min} = 10 \cdot 60 = 600$ s

↓

3. ENERGY
$W = Pt = 25 \cdot 600 = 15{,}000\ \text{J} = 15\ \text{kJ}$

Worked: energy absorbed by $10\,\Omega$ with $i(t) = 2e^{-t}$ A, $0 \to \infty$

1. POWER FORMULA
$p(t) = i(t)^2\,R$

↓

2. SQUARE THE CURRENT
$\left(2e^{-t}\right)^2 = 2^2 \cdot \left(e^{-t}\right)^2 = 4e^{-2t}$ *(square each factor, double the exponent)*

↓

3. MULTIPLY BY R
$p(t) = 4e^{-2t} \cdot 10 = 40e^{-2t}$ W

↓

4. SET UP THE INTEGRAL
$w = \lim_{b\to\infty} \int_0^b 40e^{-2t}\,dt$

↓

5. ANTIDERIVATIVE
$F(t) = \frac{40}{-2}e^{-2t} = -20e^{-2t}$

↓

6. EVALUATE AT THE TOP ($t = b$)
$F(b) = -20e^{-2b}$

↓

7. LET $b \to \infty$
$e^{-2b} \to 0$, so $F(b) \to 0$

↓

8. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -20e^0 = -20$

↓

9. TOP MINUS BOTTOM
$w = 0 - (-20) = 20$ J

↓

10. CHECK WITH THE SHORTCUT
$\frac{A}{a} = \frac{40}{2} = 20$ ✓

Worked: (definition, then Ohm) $q(t) = 3t^2$ mC through 2 kΩ. v and p at $t = 1$ s

1. DEFINITION: CHARGE TO CURRENT
$i(t) = \dfrac{dq}{dt} = 6t$ mA. $i(1) = 6$ mA $= 0.006$ A

↓

2. OHM: CURRENT TO VOLTAGE
$v = iR = 0.006 \cdot 2000 = 12$ V

↓

3. POWER ABSORBED
$p = i^2R = (0.006)^2 \cdot 2000 = 0.000036 \cdot 2000 = 0.072$ W $= 72$ mW

↓

4. CHECK
$p = vi = 12 \cdot 0.006 = 0.072$ W ✓

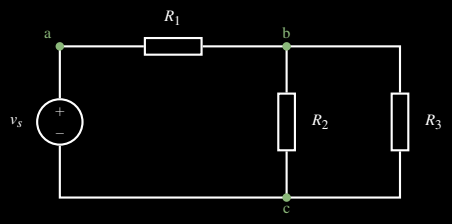
11. nodes, branches, loops

- **Branch**: one element (a source or a resistor).
- **Node**: a point where two or more branches connect.
- **Loop**: a closed path that passes each node one time only.

For b branches, n nodes, and l independent loops: $b = l + n - 1$.

Series: elements that carry the same current. **Parallel**: elements across the same two nodes, so they have the same voltage.

All points joined by plain wire are one node. Merge them before you count.



Worked: count the circuit above

1. BRANCHES
v_s, R_1, R_2, R_3 ⇒ $b = 4$

↓

2. NODES
a, b, c *(the top wire from R_2 to R_3 is one node b)* ⇒ $n = 3$

↓

3. LOOPS FROM THE FORMULA
$l = b - n + 1 = 4 - 3 + 1 = 2$

↓

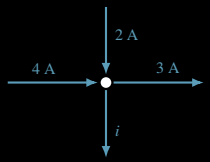
4. CHECK BY LOOKING
Loop 1: v_s, R_1, R_2. Loop 2: R_2, R_3. ✓
R_2 and R_3 share nodes b and c: they are in parallel.

12. Kirchhoff's current law (KCL)

Charge cannot collect at a node. So the current in equals the current out.

$$\sum i_{\text{in}} = \sum i_{\text{out}} \qquad \text{or} \qquad \sum_{k=1}^{N} i_k = 0$$

Sign rule for the second form: into the node is $+$, out of the node is $-$.



Worked: find i in the figure

1. LIST THE CURRENTS IN
4 A and 2 A

↓

2. LIST THE CURRENTS OUT
3 A and i

↓

3. IN EQUALS OUT
$4 + 2 = 3 + i$

↓

4. SOLVE
$6 = 3 + i \Rightarrow i = 6 - 3 = 3$ A

Worked: (negative answer) 1 A in; 3 A and i out

1. IN EQUALS OUT
$1 = 3 + i$

↓

2. SOLVE
$i = 1 - 3 = -2$ A

↓

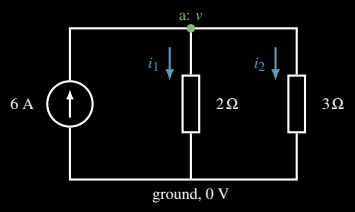
3. READ THE SIGN
i is negative, so the real current is 2 A into the node, against the arrow.

KCL in a full circuit — the procedure

1. Label each unknown current with an arrow. Guess the direction. A negative answer means the real current flows the other way.
2. Pick one node as **ground** (0 V). Name the voltage at each other node.
3. Write each resistor current with Ohm's law: $i = \dfrac{v_{\text{from}} - v_{\text{to}}}{R}$. Current flows from the higher voltage to the lower voltage.
4. Write KCL at each node except ground. A circuit with n nodes gives $n - 1$ independent KCL equations.
5. Solve. Then check KCL at each node with the numbers.

Closed surface: KCL holds for any closed boundary around several nodes. The current into the boundary equals the current out.

Current sources in parallel add, with their directions. Two different current sources in series break KCL.



Worked: the circuit above. Find v, i_1, and i_2

1. NAME THE NODE VOLTAGE
Bottom wire = ground, 0 V. Top node a = v.

↓

2. EACH RESISTOR CURRENT (OHM)
$i_1 = \dfrac{v-0}{2} = \dfrac{v}{2}, \quad i_2 = \dfrac{v-0}{3} = \dfrac{v}{3}$

↓

3. KCL AT NODE a
In: 6 A. Out: $i_1 + i_2$. $\quad 6 = \dfrac{v}{2} + \dfrac{v}{3}$

↓

4. COMMON DENOMINATOR
$\dfrac{v}{2} + \dfrac{v}{3} = \dfrac{3v}{6} + \dfrac{2v}{6} = \dfrac{5v}{6}$

↓

5. SOLVE FOR v
$\dfrac{5v}{6} = 6 \Rightarrow 5v = 36 \Rightarrow v = 7.2$ V

↓

6. EACH CURRENT
$i_1 = \dfrac{7.2}{2} = 3.6$ A, $\quad i_2 = \dfrac{7.2}{3} = 2.4$ A

↓

7. CHECK
KCL: $3.6 + 2.4 = 6$ ✓ Current divider: $\dfrac{3}{2+3} \cdot 6 = 3.6$ ✓

Worked: (dependent source) 10 A into a node. Out: i_x through $4\,\Omega$ to ground, and a CCCS of $1.5\,i_x$

1. KCL
In = out: $10 = i_x + 1.5\,i_x$

↓

2. COMBINE LIKE TERMS
$10 = 2.5\,i_x$

↓

3. SOLVE
$i_x = \dfrac{10}{2.5} = 4$ A

↓

4. DEPENDENT SOURCE CURRENT
$1.5 \cdot 4 = 6$ A

↓

5. NODE VOLTAGE (OHM ON THE $4\,\Omega$)
$v = 4 \cdot 4 = 16$ V

↓

6. CHECK
$4 + 6 = 10$ ✓

Worked: (closed surface) a boundary around two nodes

Currents that cross the boundary: 5 A in, 2 A out, 1 A in, and i out.

1. IN EQUALS OUT
$5 + 1 = 2 + i$

↓

2. SOLVE
$6 = 2 + i \Rightarrow i = 4$ A *(the currents inside the boundary do not matter)*

Which law? — KCL, KVL, or Ohm

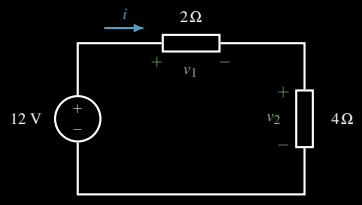
Unknown	Use
one current at a node, the others known	KCL at that node
one voltage in a loop, the others known	KVL around that loop
v or i of one resistor, the other known	Ohm's law
both v and i unknown	write KCL or KVL with Ohm's law put in, then solve

13. Kirchhoff's voltage law (KVL)

Energy is conserved, so the voltages around a closed loop add to zero.

$$\sum v = 0 \qquad \text{or} \qquad \sum v_{\text{rises}} = \sum v_{\text{drops}}$$

Sign rule: walk around the loop in one direction. If you enter an element at its + mark, write $+v$. If you enter at its − mark, write $-v$.
Pick one direction and keep it for the full loop.



Worked: the loop above. Find i, each voltage, and each power

1. WALK CLOCKWISE FROM THE BOTTOM LEFT
Source: enter at $- \Rightarrow -12$. R_1: enter at $+ \Rightarrow +v_1$. R_2: enter at $+ \Rightarrow +v_2$.

↓

2. KVL
$-12 + v_1 + v_2 = 0$

↓

3. OHM'S LAW FOR EACH RESISTOR
$v_1 = 2i, \quad v_2 = 4i$

↓

4. SUBSTITUTE
$-12 + 2i + 4i = 0$

↓

5. COMBINE LIKE TERMS
$-12 + 6i = 0$

↓

6. SOLVE FOR i
$6i = 12 \Rightarrow i = \dfrac{12}{6} = 2$ A

↓

7. EACH VOLTAGE
$v_1 = 2 \cdot 2 = 4$ V, $\quad v_2 = 4 \cdot 2 = 8$ V

↓

8. POWER ABSORBED BY EACH RESISTOR
$p_1 = i^2 R_1 = 2^2 \cdot 2 = 8$ W, $\quad p_2 = i^2 R_2 = 2^2 \cdot 4 = 16$ W

↓

9. POWER OF THE SOURCE
The arrow enters the − mark. $p = -v\,i = -(12)(2) = -24$ W (delivers 24 W)

↓

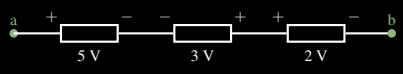
10. CHECK
KVL: $4 + 8 = 12$ ✓ Power: $-24 + 8 + 16 = 0$ ✓

Voltage between two points — the path rule

v_{ab} is the voltage of point a measured from point b. Walk any path from a to b and add the element voltages:

- You meet the + mark first: write $+v$ (a drop).
- You meet the − mark first: write $-v$ (a rise).

Every path from a to b gives the same answer. That is KVL. $v_{ba} = -v_{ab}$.
Voltage sources in series add, with their polarities: 9 V and 3 V that oppose give 6 V. Two different voltage sources in parallel break KVL.



Worked: find v_{ab} in the figure above

1. FIRST ELEMENT
You meet $+$ first: $+5$

↓

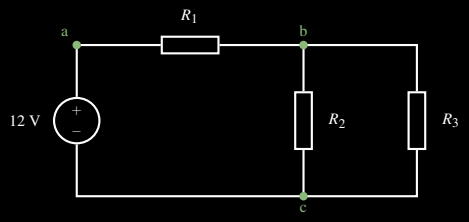
2. SECOND ELEMENT
You meet $-$ first: -3

↓

3. THIRD ELEMENT
You meet $+$ first: $+2$

↓

4. ADD
$v_{ab} = 5 - 3 + 2 = 4$ V. $\quad v_{ba} = -4$ V



Worked: (two loops) $v_s = 12$ V and $v_1 = 5$ V on R_1. Find v_2 and v_3

All resistor $+$ marks are on the top (or left) end.

1. LOOP 1: v_s, R_1, R_2
$-12 + 5 + v_2 = 0$

↓

2. SOLVE
$v_2 = 12 - 5 = 7$ V

↓

3. LOOP 2: R_2 UP, THEN R_3 DOWN
Up R_2: meet $-$ first, $-v_2$. Down R_3: meet $+$ first, $+v_3$. $\quad -v_2 + v_3 = 0$

↓

4. SOLVE
$v_3 = v_2 = 7$ V *(KVL shows that parallel elements share one voltage)*

Worked: (dependent source) one loop: 36 V source, $2\,\Omega$, $4\,\Omega$, and a CCVS of $3i$

The current i flows clockwise. It leaves the $+$ mark of the 36 V source and enters the $+$ mark of each other element.

1. KVL CLOCKWISE
$-36 + 2i + 4i + 3i = 0$ *(the CCVS voltage is $3i$, entered at its $+$ mark)*

↓

2. COMBINE LIKE TERMS
$-36 + 9i = 0$

↓

3. SOLVE
$i = \dfrac{36}{9} = 4$ A

↓

4. EACH VOLTAGE
$2 \cdot 4 = 8$ V, $\quad 4 \cdot 4 = 16$ V, $\quad$ CCVS: $3 \cdot 4 = 12$ V

↓

5. KVL CHECK
$8 + 16 + 12 = 36$ ✓

↓

6. POWER
Source delivers $36 \cdot 4 = 144$ W. Resistors absorb $4^2 \cdot 2 + 4^2 \cdot 4 = 32 + 64 = 96$ W. CCVS absorbs $12 \cdot 4 = 48$ W.

↓

7. POWER CHECK
$96 + 48 = 144$ ✓

14. series and parallel resistors

Series: one current flows through each resistor. The resistances add.

$$R_{eq} = R_1 + R_2 + \cdots + R_N$$

Parallel: one voltage is across each resistor. The conductances add.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

Two in parallel (product over sum): $R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$

N equal resistors R in parallel: $R_{eq} = \dfrac{R}{N}$

A parallel R_{eq} is smaller than the smallest resistor.

The reciprocal formula gives $\frac{1}{R_{eq}}$. Flip the result at the end.

Worked: three in parallel: 10, 20, $20\,\Omega$

1. EACH RECIPROCAL
$\dfrac{1}{10} = 0.1, \quad \dfrac{1}{20} = 0.05, \quad \dfrac{1}{20} = 0.05$

↓

2. ADD
$\dfrac{1}{R_{eq}} = 0.1 + 0.05 + 0.05 = 0.2$ S

↓

3. FLIP
$R_{eq} = \dfrac{1}{0.2} = 5\,\Omega$ *(smaller than 10 ✓)*

Worked: four $100\,\Omega$ resistors in parallel

1. EQUAL-RESISTOR RULE
$R_{eq} = \dfrac{R}{N} = \dfrac{100}{4} = 25\,\Omega$

↓

2. CHECK WITH RECIPROCALS
$4 \cdot \dfrac{1}{100} = 0.04. \quad \dfrac{1}{0.04} = 25\,\Omega$ ✓

Worked: $R_{eq} = 2 + \left[12 \parallel (4+6 \parallel 3)\right]\ \Omega$

Start at the part farthest from the terminals. Work toward the terminals.

1. INNERMOST PARALLEL PAIR $6 \parallel 3$

$$\frac{6\cdot 3}{6+3} = \frac{18}{9} = 2\,\Omega$$

↓

2. SERIES $4+2$

$$4+2 = 6\,\Omega$$

↓

3. PARALLEL $12 \parallel 6$

$$\frac{12\cdot 6}{12+6} = \frac{72}{18} = 4\,\Omega$$

↓

4. SERIES $2+4$

$$R_{eq} = 2+4 = 6\,\Omega$$

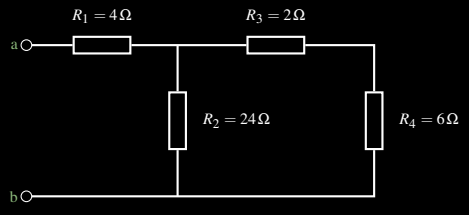
How to spot series and parallel

1. Label the nodes. All points joined by plain wire get one label.
2. **Series test**: two elements share a node, and no third element connects at that node.
3. **Parallel test**: the two ends of both elements connect to the same two nodes.
4. Combine one pair. Redraw. Repeat from the end farthest from the terminals.

A wire across a resistor shorts it. The resistor has 0 V and carries no current. Remove it: $R \parallel 0 = 0$.

A resistor with one end connected to nothing carries no current. Remove it.

R_{eq} depends on the two terminals. The same network seen from a–b and from c–d gives different values.



Worked: R_{ab} ***of the ladder above***

1. FAR END: R_3 **AND** R_4 **IN SERIES**

They share a node with nothing else on it. $2+6 = 8\,\Omega$

↓

2. PARALLEL WITH R_2

Both connect between the top node and the bottom wire. $\dfrac{24\cdot 8}{24+8} = \dfrac{192}{32} = 6\,\Omega$

↓

3. SERIES WITH R_1

$$R_{ab} = 4+6 = 10\,\Omega$$

Worked: (short) $10\,\Omega$***, then*** $6\,\Omega$ ***with a wire across it, then*** $4\,\Omega$***, all in series***

1. THE SHORTED RESISTOR

$$6 \parallel 0 = \frac{6\cdot 0}{6+0} = 0\,\Omega$$

↓

2. SERIES

$$R_{eq} = 10+0+4 = 14\,\Omega$$

Worked: (open end) $8\,\Omega$ ***from a to c,*** $5\,\Omega$ ***from c to b, and*** $7\,\Omega$ ***from c to nothing***

1. THE OPEN-ENDED RESISTOR

No closed path through the $7\,\Omega$ ⇒ no current. Remove it.

↓

2. SERIES

$$R_{ab} = 8+5 = 13\,\Omega$$

Which way? — combine or design

Given → find	Do this
the resistors → R_{eq}	combine from the far end toward the terminals
a target R_{eq} → the missing R	write the formula with R unknown, then solve

Worked: (reverse) find R ***so that*** $R \parallel 12\,\Omega = 4\,\Omega$

1. RECIPROCAL FORMULA

$$\frac{1}{4} = \frac{1}{R} + \frac{1}{12}$$

↓

2. ISOLATE $\frac{1}{R}$

$$\frac{1}{R} = \frac{1}{4} - \frac{1}{12} = \frac{3}{12} - \frac{1}{12} = \frac{2}{12} = \frac{1}{6}$$

↓

3. FLIP

$$R = 6\,\Omega$$

↓

4. CHECK

$$\frac{6\cdot 12}{6+12} = \frac{72}{18} = 4\,\Omega\ \checkmark$$

Worked: (reverse, series) what R ***in series with*** $330\,\Omega$ ***gives*** 1 $k\Omega$***?***

1. SERIES FORMULA

$$1000 = 330 + R$$

↓

2. SOLVE

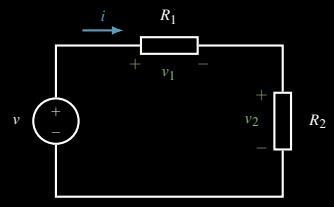
$$R = 1000 - 330 = 670\,\Omega$$

15. voltage division

Series resistors share the source voltage in proportion to their resistance:

$$v_1 = \frac{R_1}{R_1+R_2}\,v \qquad v_2 = \frac{R_2}{R_1+R_2}\,v$$

Own resistor on top. The larger resistor gets the larger voltage.



Worked: 12 ***V across*** $R_1 = 2$ $k\Omega$ ***and*** $R_2 = 4$ $k\Omega$

1. TOTAL RESISTANCE

$$R_1 + R_2 = 2+4 = 6\ \text{k}\Omega$$

↓

2. VOLTAGE ON R_1

$$v_1 = \frac{2}{6}\cdot 12 = 4\ \text{V}$$

↓

3. VOLTAGE ON R_2

$$v_2 = \frac{4}{6}\cdot 12 = 8\ \text{V}$$

↓

4. CURRENT

$$i = \frac{12\ \text{V}}{6000\,\Omega} = 0.002\ \text{A} = 2\ \text{mA}$$

↓

5. POWER ABSORBED BY EACH

$$p_1 = (0.002)^2 \cdot 2000 = 0.008\ \text{W} = 8\ \text{mW}$$
$$p_2 = (0.002)^2 \cdot 4000 = 0.016\ \text{W} = 16\ \text{mW}$$

↓

6. CHECK

$4+8 = 12$ V ✓ Source: $12\cdot 0.002 = 24$ mW $= 8+16$ ✓

Worked: (reverse) 12 V source, $R_1 = 9$ kΩ. Find R_2 so that $v_2 = 3$ V

1. DIVIDER WITH R_2 UNKNOWN
$3 = \dfrac{R_2}{9+R_2} \cdot 12$ *(kΩ throughout)*

↓

2. CLEAR THE FRACTION
$3(9+R_2) = 12R_2$

↓

3. DISTRIBUTE
$27 + 3R_2 = 12R_2$

↓

4. SOLVE
$27 = 9R_2 \Rightarrow R_2 = 3$ kΩ

↓

5. CHECK
$\dfrac{3}{9+3} \cdot 12 = \dfrac{3}{12} \cdot 12 = 3$ V ✓

More than two resistors, and voltages to ground

N resistors in series: each gets its share of the total.

$$v_k = \frac{R_k}{R_1 + R_2 + \cdots + R_N}\, v$$

Voltage at a node (measured to the bottom, ground): add the voltages of everything between that node and ground.

Worked: 24 V across $R_1 = 1$, $R_2 = 2$, $R_3 = 3$ kΩ in series (top to bottom)

1. TOTAL RESISTANCE
$1 + 2 + 3 = 6$ kΩ

↓

2. EACH VOLTAGE
$v_1 = \dfrac{1}{6} \cdot 24 = 4$ V, $\quad v_2 = \dfrac{2}{6} \cdot 24 = 8$ V, $\quad v_3 = \dfrac{3}{6} \cdot 24 = 12$ V

↓

3. CHECK
$4 + 8 + 12 = 24$ ✓

↓

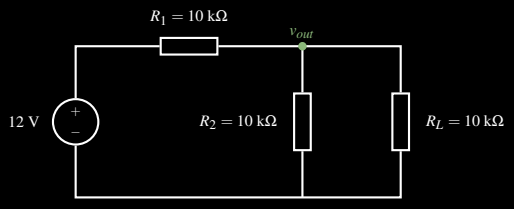
4. NODE BETWEEN R_2 AND R_3
Only R_3 is below it: $\quad 12$ V

↓

5. NODE BETWEEN R_1 AND R_2
R_2 and R_3 are below it: $\quad 8 + 12 = 20$ V

The loaded divider

A **load** R_L connected across R_2 is in parallel with R_2. It draws current, so R_1 and R_2 no longer carry the same current.
Do not use the divider on R_1 and R_2 alone when a load is connected.
Combine $R_2 \parallel R_L$ first. Then divide.



Worked: v_{out} in the figure, without and with the load

1. NO LOAD: DIVIDER
$v_{out} = \dfrac{10}{10+10} \cdot 12 = 6$ V

↓

2. WITH THE LOAD: COMBINE $R_2 \parallel R_L$
$\dfrac{10 \cdot 10}{10+10} = \dfrac{100}{20} = 5$ kΩ

↓

3. WITH THE LOAD: DIVIDER
$v_{out} = \dfrac{5}{10+5} \cdot 12 = \dfrac{5}{15} \cdot 12 = 4$ V

↓

4. READ THE RESULT
The load pulls v_{out} down from 6 V to 4 V.

16. current division

Parallel resistors share the total current i:

$$i_1 = \frac{R_2}{R_1 + R_2}\, i \qquad i_2 = \frac{R_1}{R_1 + R_2}\, i$$

The *other* resistor goes on top. The smaller resistor gets the larger current.
Which divider? Series resistors share one current, so the voltage splits (own R on top). Parallel resistors share one voltage, so the current splits (other R on top).

Worked: 6 A into 3 Ω ∥ 6 Ω

1. SUM OF THE TWO
$3 + 6 = 9\,\Omega$

↓

2. CURRENT IN THE 3 Ω
$i_1 = \dfrac{6}{9} \cdot 6 = 4$ A *(the other resistor, 6, on top)*

↓

3. CURRENT IN THE 6 Ω
$i_2 = \dfrac{3}{9} \cdot 6 = 2$ A

↓

4. CHECK
KCL: $4 + 2 = 6$ ✓ $\quad$ Same voltage: $4 \cdot 3 = 2 \cdot 6 = 12$ V ✓

More than two branches: use conductances

For N resistors in parallel, the product-over-sum form does not work. Use conductances $G = \frac{1}{R}$:

$$i_k = \frac{G_k}{G_1 + G_2 + \cdots + G_N}\, i$$

Own G on top. The largest G (the smallest R) gets the largest current.

Worked: 12 A into 2, 3, and 6 Ω in parallel

1. EACH CONDUCTANCE
$G_1 = \dfrac{1}{2} = 0.5, \quad G_2 = \dfrac{1}{3} = 0.333, \quad G_3 = \dfrac{1}{6} = 0.167$ S

↓

2. TOTAL
$0.5 + 0.333 + 0.167 = 1$ S

↓

3. EACH CURRENT
$i_1 = \dfrac{0.5}{1} \cdot 12 = 6$ A, $\quad i_2 = \dfrac{0.333}{1} \cdot 12 = 4$ A, $\quad i_3 = \dfrac{0.167}{1} \cdot 12 = 2$ A

↓

4. CHECK
KCL: $6 + 4 + 2 = 12$ ✓ $\quad$ Same voltage: $6 \cdot 2 = 4 \cdot 3 = 2 \cdot 6 = 12$ V ✓

***Worked: (reverse)** $10\,A$ **into** $R_1 = 4\,\Omega \parallel R_2$**. Find** R_2 **so that** $i_2 = 2\,A$*

1. DIVIDER WITH R_2 UNKNOWN
$$i_2 = \frac{R_1}{R_1 + R_2}\, i \Rightarrow 2 = \frac{4}{4 + R_2} \cdot 10$$

↓

2. CLEAR THE FRACTION
$$2(4 + R_2) = 40$$

↓

3. DISTRIBUTE
$$8 + 2R_2 = 40$$

↓

4. SOLVE
$$2R_2 = 32 \Rightarrow R_2 = 16\,\Omega$$

↓

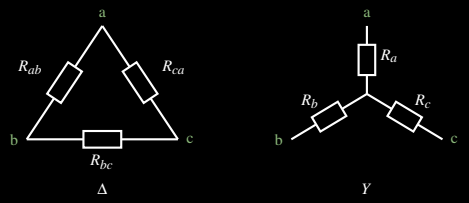
5. CHECK
$$\frac{4}{4+16} \cdot 10 = \frac{4}{20} \cdot 10 = 2\text{ A} \checkmark$$

Which divider? — the decision table

You have	Use
series resistors and their total v	voltage divider: own R on top
two parallel resistors and their total i	current divider: other R on top
N parallel resistors and their total i	current divider with G: own G on top
a load across one series resistor	combine the load first, then the voltage divider
the current through a resistor	no divider: Ohm's law, $v = iR$

17. wye–delta conversion

Some networks are neither series nor parallel (a bridge, for example). Convert a Δ (triangle) to a Y (star), or the reverse. Both connect the same three nodes a, b, c.



$\Delta \to Y$: each Y resistor $= \dfrac{\text{product of the two } \Delta \text{ sides that touch its node}}{\text{sum of all three } \Delta \text{ sides}}$

$$R_a = \frac{R_{ab}R_{ca}}{\Sigma} \quad R_b = \frac{R_{ab}R_{bc}}{\Sigma} \quad R_c = \frac{R_{bc}R_{ca}}{\Sigma}$$

$Y \to \Delta$: each Δ side $= \dfrac{N}{\text{the } Y \text{ resistor opposite it}}$, where
$N = R_aR_b + R_bR_c + R_cR_a$.

$$R_{ab} = \frac{N}{R_c} \quad R_{bc} = \frac{N}{R_a} \quad R_{ca} = \frac{N}{R_b}$$

Balanced: all three equal. Then $R_Y = \frac{R_\Delta}{3}$ and $R_\Delta = 3R_Y$.

***Worked:** $\Delta \to Y$ **with** $R_{ab} = 10$, $R_{bc} = 20$, $R_{ca} = 30\,\Omega$*

1. SUM
$$\Sigma = 10 + 20 + 30 = 60\,\Omega$$

↓

2. R_a (TOUCHES R_{ab} AND R_{ca})
$$R_a = \frac{10 \cdot 30}{60} = \frac{300}{60} = 5\,\Omega$$

↓

3. R_b (TOUCHES R_{ab} AND R_{bc})
$$R_b = \frac{10 \cdot 20}{60} = \frac{200}{60} = 3.33\,\Omega$$

↓

4. R_c (TOUCHES R_{bc} AND R_{ca})
$$R_c = \frac{20 \cdot 30}{60} = \frac{600}{60} = 10\,\Omega$$

***Worked: (reverse)** $Y \to \Delta$ **with** $R_a = 5$, $R_b = 3.33$, $R_c = 10\,\Omega$*

1. THE THREE PRODUCTS
$R_aR_b = 5 \cdot 3.33 = 16.67$
$R_bR_c = 3.33 \cdot 10 = 33.33$
$R_cR_a = 10 \cdot 5 = 50$

↓

2. ADD THEM
$$N = 16.67 + 33.33 + 50 = 100$$

↓

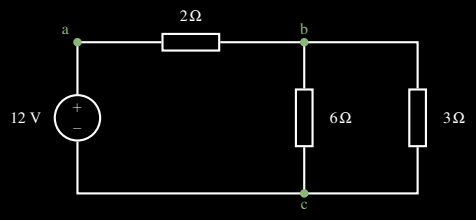
3. EACH SIDE OVER THE OPPOSITE Y
$$R_{ab} = \frac{100}{R_c} = \frac{100}{10} = 10\,\Omega$$
$$R_{bc} = \frac{100}{R_a} = \frac{100}{5} = 20\,\Omega$$
$$R_{ca} = \frac{100}{R_b} = \frac{100}{3.33} = 30\,\Omega$$

↓

4. CHECK
The same 10, 20, 30 as the example above ✓

18. putting it together: one full circuit

This uses sections 10 to 16 in order: reduce, find the source current, then expand back out.



Worked: find every current, voltage, and power

1. PARALLEL PAIR $R_2 \parallel R_3$
$$\frac{6 \cdot 3}{6+3} = \frac{18}{9} = 2\,\Omega$$

↓

2. SERIES WITH R_1
$$R_{eq} = 2 + 2 = 4\,\Omega$$

↓

3. SOURCE CURRENT (OHM ON R_{eq})
$$i = \frac{12}{4} = 3\text{ A} \quad \textit{(this current flows through } R_1\textit{)}$$

↓

4. VOLTAGE ON R_1
$$v_1 = iR_1 = 3 \cdot 2 = 6\text{ V}$$

↓

5. VOLTAGE ON THE PARALLEL PAIR (KVL)
$$v_{bc} = 12 - 6 = 6\text{ V} \quad \textit{(same on } R_2 \textit{ and } R_3\textit{)}$$

↓

6. CURRENT IN EACH BRANCH (OHM)
$$i_2 = \frac{6}{6} = 1\text{ A}, \quad i_3 = \frac{6}{3} = 2\text{ A}$$

↓

7. KCL CHECK AT NODE b
in 3 A = out $1 + 2$ A ✓

↓

8. POWER ABSORBED BY EACH RESISTOR
$$p_1 = 3^2 \cdot 2 = 18\text{ W}, \quad p_2 = 1^2 \cdot 6 = 6\text{ W}, \quad p_3 = 2^2 \cdot 3 = 12\text{ W}$$

↓

9. POWER CHECK
Source delivers $12 \cdot 3 = 36$ W. Absorbed: $18 + 6 + 12 = 36$ W ✓